

Question 14

Q: A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.

Theoretical Concept: Satellite Image Processing

In satellite imaging, **spatial resolution** (sampling) determines if we can see individual buildings vs city blocks. **Spectral resolution** (the number of color/infrared channels acquired) determines if we can differentiate vegetation types.

Proper acquisition sensors (multispectral scanners) are vital; if the sensor is inadequate, no amount of digital enhancement can reliably classify land structures.

OpenCV Python Solution

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```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical mat
import matplotlib.pyplot as plt # Import Matplotlib to visualize th

# Step 1: Simulate the acquisition of Red and Near-Infrared (NIR) satellite bands
red_band = np.full((100, 100), 50, dtype=np.float32) # Create a solid gray image array filled w
nir_band = np.full((100, 100), 150, dtype=np.float32) # Create a solid gray image array filled

# Step 2: Introduce a "forest" area in the center (High NIR reflection, Low Red reflection)
cv2.rectangle(red_band, (30, 30), (70, 70), 20, -1) # Draw a filled or outlined rectangle on th
cv2.rectangle(nir_band, (30, 30), (70, 70), 220, -1) # Draw a filled or outlined rectangle on t

# Step 3: Calculate the Normalized Difference Vegetation Index (NDVI)
# Formula: (NIR - Red) / (NIR + Red)
ndvi = (nir_band - red_band) / (nir_band + red_band) # Calculate Normalized Difference Vegetati

# Step 4: Display the vegetation map
plt.imshow(ndvi, cmap='RdYlGn') # Render the image array to the display
plt.title('Calculated NDVI (Vegetation Index)') # Add a descriptive title to the plot
```

```
plt.colorbar() # Add a color legend bar to the plot
plt.show() # Show the final plot window to the user
```

Expected Output

The output renders a topographical color map. The synthetic 'forest' area clearly highlights in dark green because its mathematical NDVI score approaches +1.0, a standard metric in satellite CV.

How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](https://colab.research.google.com/) and create a New Notebook.
- Click the  **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File > Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.